ADCG: A Cross-Modality Domain Transfer Learning Method for Synthetic Aperture Radar in Ship Automatic Target Recognition

Gui Gao, *Member, IEEE*, Yuxi Dai, Xi Zhang, Dingfeng Duan, and Fei Guo

***Abstract*— Due to the powerful feature extraction and expression ability of convolutional neural networks (CNNs), exceptional success has been achieved in the field of ship automatic target recognition (ATR) of synthetic aperture radar (SAR). However, the CNNs cannot work effectively with sparse labeled samples and imbalanced categories. This study proposes a new attention-dense-CycleGAN (ADCG) method that is suitable for the ship transfer learning task from optical to SAR (OPT2SAR). The key improvement of the ADCG lies in the construction of a dense connection module (DCM) and a lightweight convolutional block attention module (CBAM). The DCM is able to overcome the problems of generator feature redundancy, large network model parameters, and severe training time in the original CycleGAN network. The lightweight CBAM can solve the problem of not being able to locate the main features of ships with a minimal increase in network parameters. Compared with the performance of other popular generative adversarial networks, the superior performance of the ADCG in the OPT2SAR transfer learning is demonstrated with the Fréchet inception distance (FID) minimum of 76.04 and the Kernel inception distance (KID) minimum of 0.0403. Finally, the ability of pseudo-SAR domain images was tested to improve the recognition accuracy of popular ship classification networks, and this achieved an average improvement of 6% in recognition accuracy. Therefore, the results of this study verify the rationality, validity, and application value of pseudo-SAR domain in solving the problems of sparse marker samples and class imbalance in ship ATR network model.**

***Index Terms*— Generative adversarial networks (GANs), image-to-image translation (I2IT), optical to SAR (OPT2SAR), ship classification, synthetic aperture radar (SAR).**

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I. Introduction

SYNTHETIC aperture radar (SAR) and optical sensors with different imaging capabilities are currently widely used as earth observation sensors. Optical images are characterized by a large sample size owing to their simple data acquisition approach, ease of understanding, and labeling of image content. SAR, as an active microwave remote sensing technology, is unaffected by climatic and natural environmental factors. It can penetrate clouds in all-weather conditions and has the advantage of obtaining stable and continuous surface information for a long time. Therefore, SAR is widely used in each application in various fields [1]. The ship automatic target recognition (ATR) technology of SAR images is an important branch of remote sensing image interpretation, and it is also of great interest because of its importance in marine monitoring, marine transportation management, and so on [2], [3], [4], [5].

There are three main traditional methods for SAR in ship target recognition [6], such as template-, model-, and feature-based methods. The template-based method distinguishes target types by comparing the level of similarity between the target image and the template, but a lack of target image templates hinders the implementation of this method. The model-based method relies on complex scattering models to describe target images and determine target types by using the likelihood estimation of model parameters. However, the nonlinear characteristics of complex scattering models affect the accurate estimation of these parameters, and feature-based methods use a classifier to classify images after feature extraction. However, owing to the influence of SAR imaging characteristics, these features are prone to variability and have a problem of unstable feature extraction. Due to the rapid-developed method of deep learning, methods based on convolutional neural networks (CNNs), such as VGG16 [7], the ResNet series [8], DenseNet [9], MobileNet [10], InceptionNet [11], and EfficientNet [12], have achieved substantial success in ship ATR through data-driven methods based on the powerful feature expression capabilities of CNNs.

Although the SAR ship target recognition method based on CNNs is significantly effective at present, it still needs a large amount of effective labeled data of SAR images to train the classifiers. The visual information provided by SAR sensors is not as rich as those provided by optical sensors. There is a good deal of speckle noise in the SAR images, and human vision cannot adapt to the microwave scattering

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phenomena, making it difficult to understand the complex scattering characteristics. Therefore, interpreting SAR images to obtain labeled samples requires expert knowledge, and target labeling in large-scale SAR ship datasets is time-consuming. Therefore, they are characterized by a small sample size and imbalanced categories [13], [14]. Under the current situation of scarce training samples, the training task of SAR in ship ATR technology based on deep-learning network models has become extremely difficult, which has a considerable impact on the accuracy improvement of the SAR ship recognition network model.

In the advantage of not increasing the complexity of the model, the popularity of ship recognition tasks based on transfer learning has gradually increased to improve recognition accuracy. For example, Wang et al. [15] completed the task of cross-domain knowledge transfer from the domain of natural image to the domain of SAR image based on the PASCAL VOC dataset. Rostami et al. [16] realized the task of cross-domain knowledge transfer from the optical remote sensing images to the SAR images. Huang et al. [17] studied the transfer learning of the maximum mean difference (MMD) from multiple fields, including the task of optical to SAR (OPT2SAR). In order to save the problem of lack of SAR labeled data, Shi et al. [18] proposed a new framework called unsupervised domain adaptation (UDA), which is based on progressive transfer for SAR ship detection tasks. Song et al. [19] combined adversarial learning and metric learning methods and then proposed a new domain transfer network called domain transfer using adversarial learning and metric learning (DTAM), which achieved military and civilian ship recognition tasks by domain adapting to intermediate and target SAR domains. Therefore, using the transfer learning method to realize the joint analysis of optical and SAR ship data and carrying out domain knowledge transfer can effectively solve the problem of scarce effective marking samples and the imbalance of categories in the SAR of ship ATR deep learning network model and improve the accuracy of the ship ATR model [20], [21], [22].

Unfortunately, cross-modality domains, such as optical and SAR, have significant differences in data distribution between the source and target domains, which may not be transmitted to the lower layers, leading to serious domain mismatch issues [17]. Some scholars overcome the difficulty of domain mismatches by aligning the data in the source and target domains and ensuring that the two domains have similar probability distributions [23], [24], [25], [26]. Other investigations proposed methods for cross-modality domain transfer by establishing mappings [27], [28]. Through cross-domain mapping, data in the source domain obtain the features of the target domain, thereby reducing the nonlinear differences between heterogeneous images. This method is called image-to-image translation (I2IT) [27], [28], [29], [30], [31], [32]. Therefore, this study aimed to work out the challenges of a limited number of effectively labeled SAR images and imbalanced categories by transferring useful knowledge from the domain of optical. Owing to the significant differences in data distribution between different cross-modality domains, current optical ship image recognition methods cannot be directly applied to SAR images. Therefore, various limitations must be overcome, and these include the following.

1) Unlike optical images, SAR images do not contain color or brightness information but mainly include the information of structure, texture, and shape [34].
2) Although feature alignment methods can reduce the discreteness of features, they are not suitable for transmission in low-level shared feature spaces because of the differences in the optical and SAR imaging mechanisms and significant domain differences [35].

Owing to the various challenges mentioned above, the task of OPT2SAR cross-modality domain knowledge transfer is challenging. However, the emergence of I2IT, a method based on the generative adversarial network (GAN) [33], provides new solutions for image translation from the OPT2SAR domain. However, most current GAN-based image translation methods, such as conditional generative adversarial networks (cGAN) [36] and Pix2Pix [27], are restricted by supervised learning and conditions that require paired training samples. In the ship recognition field, obtaining paired optical and remote sensing image data is difficult. Unsupervised image translation models, such as CycleGAN [28], U-GAT-IT [37], DualGAN [30], and DiscoGAN [29], have the advantage of not need paired images, thereby reducing the difficulty of obtaining ship slice data. CycleGAN, the most famous method in unsupervised unpaired image translation tasks, introduces the loss of cycle consistency and identity into the loss function of the original GAN, such that the image generated by one cycle has a high degree of similarity with the original image and shows an excellent translation effect in unsupervised unpaired image translation tasks.

The above adversarial learning method based on the CycleGAN network provides feasible insight into ship OPT2SAR cross-modality domain knowledge transfer task. However, in the ship image translation task of OPT2SAR, the CycleGAN network exhibited some problems.

1) The complex shore in the optical dataset interferes with the fast and accurate extraction of ship main features by the CycleGAN network. Moreover, the proportion of ship targets in the image between the optical and SAR datasets is different, and the CycleGAN network cannot overcome the problem of different scales of ship subjects in different domains during image translation.
2) The CycleGAN generator network, composed of nine stacked residual blocks (RBs), has a complex structure and many parameters, consumes considerable training time, and can also lead to feature redundancy.

In the current two-stage ship ATR transfer learning method, the first stage often directly uses CycleGAN for pixel-level domain knowledge migration, without considering the above problems and making network improvements for CycleGAN in ship ATR; therefore, the effect often has room for improvement. In this study, to solve the difficulties of CycleGAN in the knowledge transfer of the ship pixel-level domain, the uneven distribution of data categories in the ship ATR task, and the problem of insufficient data volume, CycleGAN was used as the baseline model, and it proposed a simple, fast,

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and better pixel-level cross-modality domain feature transformation method and built a cross-modality domain transfer learning framework for the automatic recognition of SAR ship targets. The main contributions of this proposed method are as follows.

1) Considering the differences in optical and SAR imaging mechanisms, a new OPT2SAR ship image translation network, attention-dense-CycleGAN (ADCG) based on CycleGAN that is suitable for the ship OPT2SAR cross-modality domain knowledge transfer tasks was proposed. A lightweight generator with a dense connection module (DCM) and a convolutional block attention module (CBAM) was constructed to generate high-quality pseudo-SAR images.
2) Based on the ADCG network, a cross-modality domain transfer learning framework for automatic target recognition (ATR) of SAR in ships was constructed, as it appears to be that this is the first work to settle the difficulty of scarcity of effectively labeled samples, imbalance of categories, and improve the precision of SAR ship ATR model by using the pseudo-SAR domain images.

Section II of this study narrates the ship ATR method based on cross-modality domain transfer learning. Section III introduces the ablation experiment and result analysis of the ADCG based on pixel-level domain transfer, as well as the classification application experiment and the result of the pseudo-SAR image analysis. Finally, the conclusions are presented in Section IV.

II. Proposed Cross-Modality Domain Transfer Learning Method

This section introduces the technical details of the cross-modality domain transfer learning framework for ship ATR task of SAR images. The overall framework of the proposed solution is illustrated in Fig. 1.

A. Overall Framework for Cross-Modality Domain Transfer Learning

Based on the ADCG network, a cross-modality domain transfer learning framework was constructed for the automatic recognition of SAR ship targets, which was implemented in two steps.

1) Using the pixel-level OPT2SAR feature transformation method of the ADCG network, cross-modality domain knowledge transferred from optical images to SAR images was achieved, constructing high-quality pseudo-SAR domains, balancing the categories of real SAR ship datasets, and expanding the data volume.
2) Utilizing pseudo-SAR domain images to assist popular ship ATR classifiers in achieving high-precision recognition of SAR ship target categories.

In the cross-modality domain ship ATR task, the optical and SAR domain ship image was used as the source and target domain data, respectively. The data of target domain can be divided into two parts: training data of the ADCG network and testing data of the popular ship classification network. Our method aims to achieve OPT2SAR cross-modality domain transfer through pixel-level feature translation, achieving the goal of balancing the categories of SAR ship datasets and expanding the data volume. ATR classifiers were used to achieve high-precision recognition of SAR ship target categories. The following provides detailed information on pixel-level feature translation.

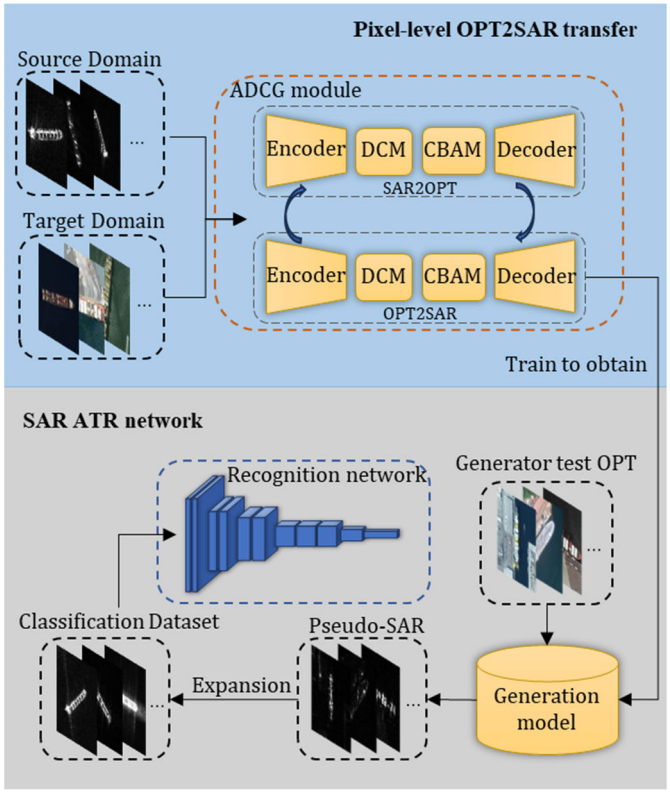


Fig. 1. Framework of OPT2SAR cross-modality domain transfer learning.

B. Pixel-Level Feature Translation Method ADCG

Owing to the different imaging mechanisms of the sensor, there is a considerable difference in the appearance of their images. To reduce the difference between the domains of ship images, the unsupervised unpaired pixel-level adversarial learning method, CycleGAN network, was utilized as the baseline network, a feature converter with DCM and CBAM, and generated a pseudo-SAR domain that could not be distinguished by the discriminator was proposed. This domain remains unchanged from the optical domain in terms of ship position, structure, shape, size, and other aspects, thereby reducing the feature differences between ship images in the SAR domain and real images. Therefore, the current popular ship ATR classifier can be extended to the real SAR domain based on the pseudo-SAR domain.

1) Baseline Network CycleGAN: A CycleGAN composed of generators and discriminators was introduced as the baseline network, replacing the stacked RBs with the proposed attention mechanism of dense connection blocks to generate pseudo-SAR domains. The loss function principle of the CycleGAN network is as follows.

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a) Basic principle of GAN: Generators need to train and learn the real data distribution while generating random samples that are similar to the real data. The discriminator's job is to distinguish between the fake and the real data. Their minimax game forms are expressed as follows:

$$\min_G \max_D \; V(D,G) = E_{v\sim P_r(v)}\left[\log D(v)\right] + E_{z\sim P_g(z)}\left[\log(1-D(G(z)))\right] \quad (1)$$

where G, D, and E stand for the generator, the discriminator, and the expectation, respectively. $v \sim P_r(v)$ on behalf of real data v and data distribution $P_r(v)$. $z \sim P_g(z)$ on behalf of the pseudo data z and its specific probability distribution $P_g(z)$. $\min_G \max_D V(D,G)$ represents the max–min game between the generator and the discriminator.

b) GAN loss: The generator generates false samples with an approximate real data distribution to deceive the discriminator during the max–min game, whereas the discriminator enhances its level to distinguish false samples while guiding the generator to synthesize more realistic false samples during confrontation with the generator. Taking the OPT2SAR image translation task as an example, assuming x_{SAR} and x_{opt} are the samples from the training dataset $\{x_{\mathrm{SAR}} \sim X_{\mathrm{SAR}}\}\{x_{\mathrm{opt}} \sim X_{\mathrm{opt}}\}$ in the optical and SAR domains, respectively, and the image translation task is aim to establish a mapping between X_{SAR} and X_{opt}. $G_{O\to S}$ stands for OPT2SAR generator, and D_{SAR} represents the discriminator of real SAR images and pseudo-SAR images. The OPT2SAR process formed by training this adversarial game under the guidance of the GAN losses is described as follows:

$$\begin{aligned} L_{\mathrm{GAN}}&\left(G_{O\to S}, D_{\mathrm{SAR}}, X_{\mathrm{opt}}, X_{\mathrm{SAR}}\right) \\ &= E_{x_{\mathrm{SAR}}\sim X_{\mathrm{SAR}}}\left[\log D_{\mathrm{SAR}}(x_{\mathrm{SAR}})\right] \\ &\quad + E_{x_{\mathrm{opt}}\sim X_{\mathrm{opt}}}\left[\log\left(1-D_{\mathrm{SAR}}\left(G_{O\to S}\left(x_{\mathrm{opt}}\right)\right)\right)\right]. \end{aligned} \quad (2)$$

c) Cycle consistency loss: Although relying solely on the GAN loss can enable the generator to have the ability to extract and generate data from x_{SAR}, it cannot guarantee the overall arrangement of the image, position, and attitude of the ship target. Therefore, to achieve translation between unpaired images, the CycleGAN network uses a cycle consistency constraint [28], [29], [30] and constructs a new generator, $G_{S\to O}$, and discriminator, D_{opt}, to form the SAR to optical (SAR2OPT) process, which uses the same GAN loss, $L_{\mathrm{GAN}}(G_{S\to O}, D_{\mathrm{opt}}, X_{\mathrm{opt}}, X_{\mathrm{SAR}})$, as the generator and discriminator in the SAR2OPT process. Finally, the two processes are combined into a closed loop to achieve a complete loop. The cycle consistency constraint can ensure that the ship image generated during a cycle remains almost unchanged in terms of overall arrangement and position attitude from the original ship image, i.e., $G_{S\to O}(G_{O\to S}(x_{\mathrm{opt}})) \approx x_{\mathrm{opt}}$ and $G_{O\to S}(G_{S\to O}(x_{\mathrm{SAR}})) \approx x_{\mathrm{SAR}}$. The cycle consistency loss function is interpreted as follows:

$$\begin{aligned} L_{\mathrm{cycle}}&\left(G_{O\to S}, G_{S\to O}, X_{\mathrm{opt}}, X_{\mathrm{SAR}}\right) \\ &= E_{x_{\mathrm{opt}}\sim X_{\mathrm{opt}}}\left[\left|\left|G_{S\to O}\left(G_{O\to S}\left(x_{\mathrm{opt}}\right)\right)-x_{\mathrm{opt}}\right|\right|_1\right] \\ &\quad + E_{x_{\mathrm{SAR}}\sim X_{\mathrm{SAR}}}\left[||G_{O\to S}(G_{S\to O}(x_{\mathrm{SAR}}))-x_{\mathrm{SAR}}||_1\right]. \end{aligned} \quad (3)$$

d) Identity loss: To ensure continuity and consistency of the generated image when mapping from the target domain, CycleGAN uses the following definition of identity loss:

$$\begin{aligned} L_{\mathrm{identity}}&\left(G_{O\to S}, G_{S\to O}, X_{\mathrm{opt}}, X_{\mathrm{SAR}}\right) \\ &= E_{x_{\mathrm{SAR}}\sim X_{\mathrm{SAR}}}\left[||G_{S\to O}(x_{\mathrm{SAR}})-x_{\mathrm{SAR}}||_1\right] \\ &\quad + E_{x_{\mathrm{opt}}\sim X_{\mathrm{opt}}}\left[\left|\left|G_{O\to S}\left(x_{\mathrm{opt}}\right)-x_{\mathrm{opt}}\right|\right|_1\right]. \end{aligned} \quad (4)$$

The final loss function, L_{OPT2SAR}, of OPT2SAR task is as follows, where α, β, and γ represent the weight:

$$L_{\mathrm{OPT2SAR}} = \alpha L_{\mathrm{GAN}} + \beta L_{\mathrm{cycle}} + \gamma L_{\mathrm{identity}}. \quad (5)$$

L_{SAR2OPT} is similar to L_{OPT2SAR} in that it has the same loss function composition and the same weight parameters.

e) Total process of CycleGAN: The CycleGAN image translation framework is shown in Fig. 2, where input S represents the SAR image and input and O represents the optical image. Pseudo O and S represent the generated pseudo-optical and pseudo-SAR images, respectively. The SAR and optical images reconstructed after one cycle are represented by the recovered $\hat{S}$ and recovered $\hat{O}$. The new image obtained by inputting $G_{S\to O}$ with the real optical image should be consistent with the original optical image, represented by the same O''. The new image obtained by inputting $G_{O\to S}$ with the real SAR image should be consistent with the original SAR image, represented by the same S''. Finally, $L_{\mathrm{cycle}(s\to o)}$, $L_{\mathrm{identity}(s\to o)}$, and $L_{\mathrm{GAN}(s\to o)}$ represent the cycle consistency, identity, and GAN losses of the SAR2OPT process, respectively. Similarly, $L_{\mathrm{cycle}(o\to s)}$, $L_{\mathrm{identity}(o\to s)}$, and $L_{\mathrm{GAN}(o\to s)}$ represent the cycle consistency, identity, and GAN losses of the OPT2SAR process, respectively.

2) Attention Mechanism Dense Connection Generator: Owing to the considerable differences between the images of optical and SAR domain, it was difficult for the original CycleGAN network to generate satisfactory images. The main information for target identification of SAR images is usually contained in the low-level feature space. The CycleGAN generator comprises of nine stacked RBs with complex network models, large parameter quantities, long training times, and easy feature redundancy. The shore-based background of the source-domain opt image is complex, whereas CycleGAN does not locate the main ship features.

To ensure that the generator can generate more convincing pseudo-SAR images, a densely connected generator was designed with an attention mechanism. The generator architecture was improved based on the original CycleGAN [28], as shown in Fig. 3. The generator included encoding, decoding, and attention-dense connection feature converters. $G_{O\to S}$ and $G_{S\to O}$ have the same network architecture but different network parameters.

1) First, the encoder is composed of one convolutional layer with a kernel size of 7, a step size of 1, two downsampling convolutional layers with a kernel size of 3, and a step size of 2 to compress and encode the image spatially.
2) The feature converter is then composed of a DCM and lightweight CBAM, which locates the main features

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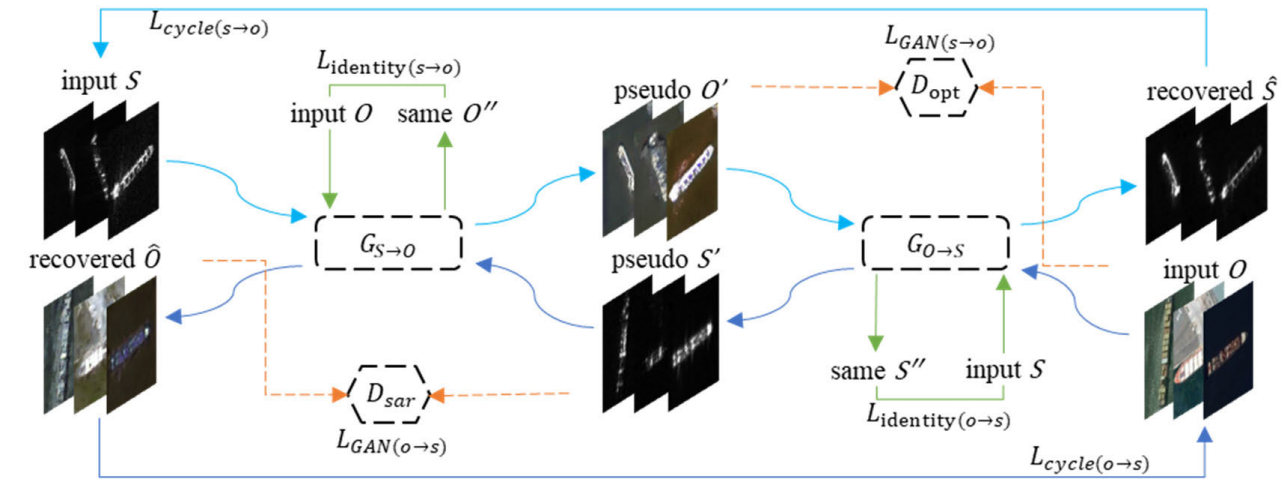


Fig. 2. Image translation framework of CycleGAN.

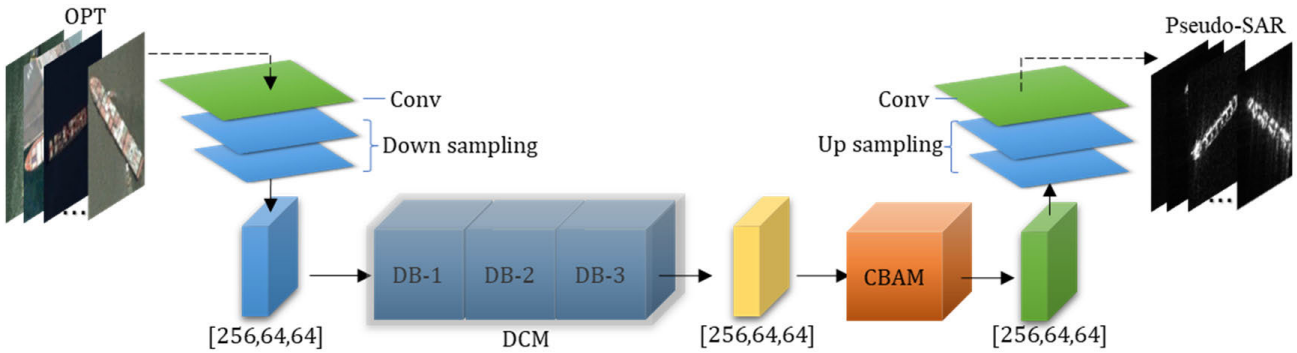


Fig. 3. Generator structure framework for the ADCG networks.

of the ship to transmit the feature map and retain its advanced features.

3) Finally, two upsampling convolutional layers with a kernel size of 3 and step size of 2 and one convolutional layer with a kernel size of seven and and step size of 1 were used to construct a decoder for reconstructing and generating the clear images.

The overall structure of the generator is simple, with fewer network parameters compared to the original CycleGAN, and avoids the feature redundancy caused by RBs. It can also locate the main ship features.

3) DCM: The detailed structure of DCM is shown in Fig. 4. The DCM of the ADCG comprises of three dense blocks (DBs), each of which is internally composed of three bottleneck blocks. The first DB was slightly different from the last two and comprises of a basic block and two bottleneck blocks.

1) The function of the basic block is to reduce the number of feature channels from 256 to 128 using a convolutional (Conv) layer.
2) The function of the bottleneck block is to use 1 × 1 Conv and 3 × 3 Conv layers to convolute the number of channels to 16 and uses the Concat operation to concatenate feature channels before and after Conv. The final feature map can maintain the same number of feature channels (256) after passing through the three DBs.

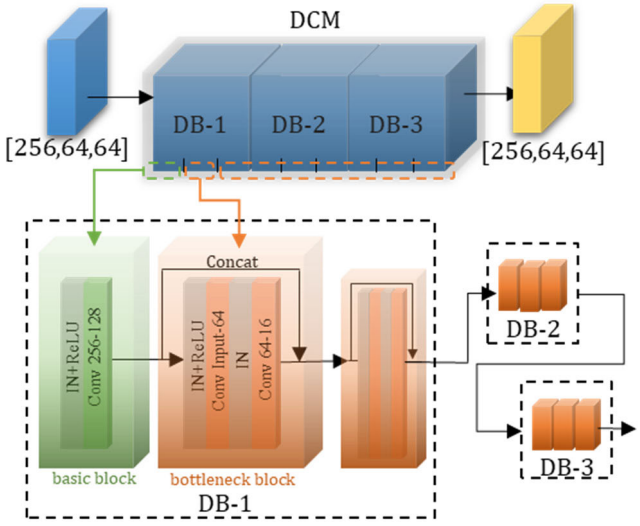


Fig. 4. Detailed structure of DCM.

The DCM reduces the number of feature channels and network computation substantially, while ensuring the quality of feature extraction. This reduces the number of feature maps but increases the reuse rate of these maps to reduce the number

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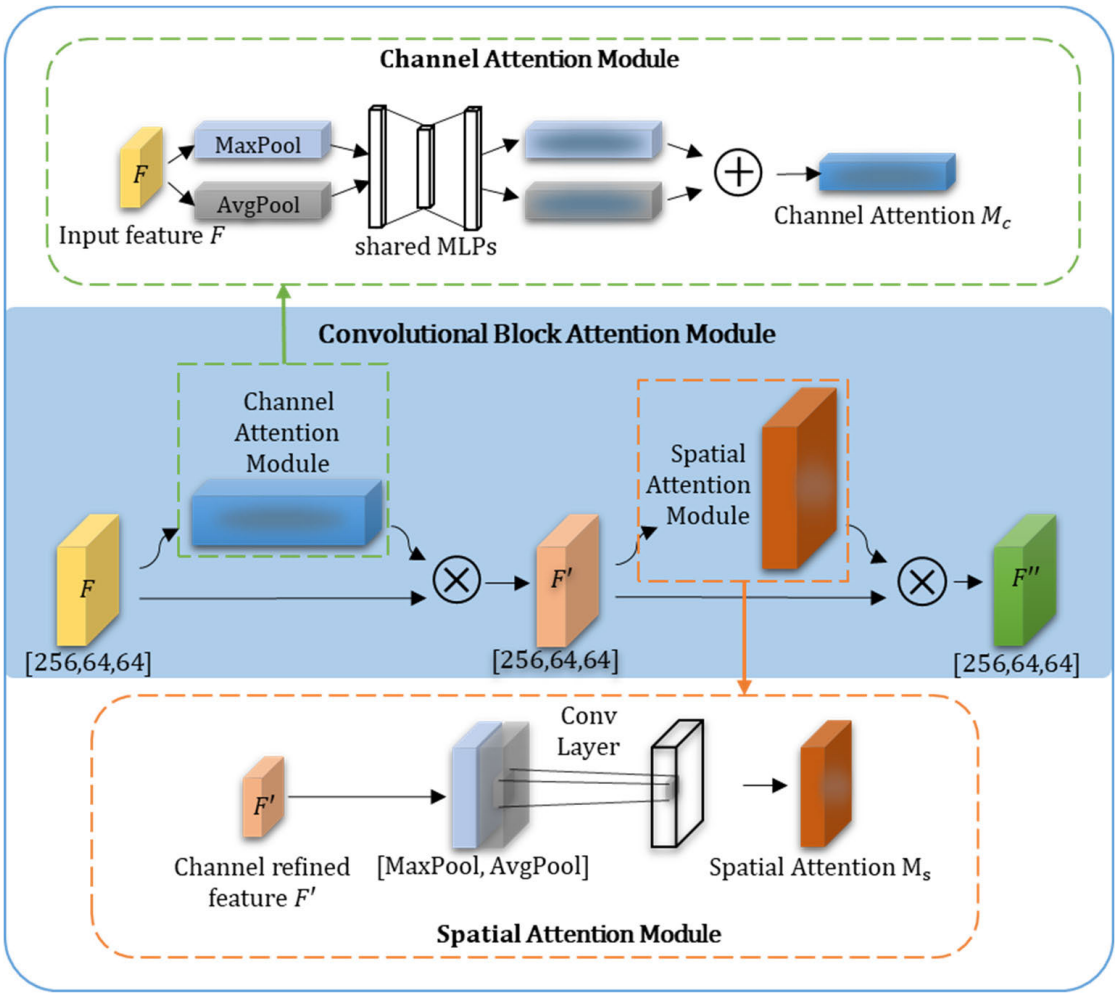


Fig. 5. Detailed structural framework diagram of CBAM and its various submodules.

of network parameters, which solves the problem of feature redundancy caused by the original CycleGAN residual block generator and improves the training speed of the network model considerably.

4) CBAM: The detailed structure of the CBAM is shown in Fig. 5. The channel attention module (CAM) and the spatial attention module (SAM) are the two submodules of the CBAM.

$F \in \mathbb{R}^{C\times H\times W}$ represents the feature map that is input into the CBAM module, and then, CBAM sequentially performs channel and spatial attention calculations to generate a 1-D channel attention map $M_c \in \mathbb{R}^{C\times 1\times 1}$ and a 2-D spatial attention map $M_s \in \mathbb{R}^{1\times H\times W}$.

The CBAM can adaptively refine intermediate feature maps, thereby ensuring computational power and parameter savings. The entire process can be summarized as follows:

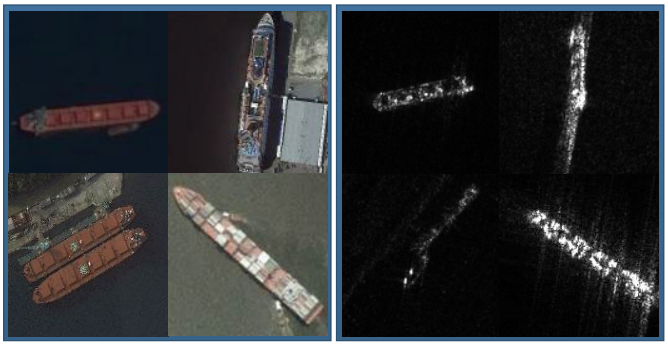
$$\begin{aligned} F' &= M_c(F) \otimes F \\ F'' &= M_s\left(F'\right) \otimes F' \end{aligned} \tag{6}$$

where $\otimes$ represents the element-by-element multiplication. F represents the input features, F' represents the features processed by the CAM module, and F'' represents those processed by the SAM module. Fig. 5 shows the detailed structure of each submodules.

1) *CAM:* So as to promote the representation ability of the network, the CAM module used both average and maximum pooling operations to combine the feature information. The generated features are forwarded to a shared network, which is composed of multilayer perceptron's (MLPs), to generate a channel attention map $M_c \in \mathbb{R}^{C\times 1\times 1}$. $\oplus$ represents the element-by-element summation and the output feature vectors are combined by this method.
2) *SAM:* The SAM has the ability to mine feature spatial relationships and generate spatial attention maps. This module uses average and maximum pooling methods to mine the important information of the feature map in the direction of the channel axis and highlight the information regions effectively. Then, the feature descriptions of the two are concatenated, convolutional layers are applied to generate spatial attention mapping $M_s \in \mathbb{R}^{1\times H\times W}$, and the emphasized or suppressed regions are encoded.

The CBAM adaptively refines the intermediate feature maps extracted by the feature converter by successively utilizing

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(a) (b)

Fig. 6. Example of OPT and SAR dataset images. (a) OPT images. (b) SAR images.

channel and position attention mechanisms while achieving the function of locating the main ship features and maintaining a lightweight number of network parameters without imposing a considerable burden on the network model's training time.

III. Experimental Results and Analysis

Section III summarizes the experimental process and provides information on the experimental details.

1) Section III-A provides an overview of the database used in the experiment, details of dataset partitioning, experimental environment, parameter settings, and evaluation indicators.
2) In Section III-B, an ADCG network ablation experiment was conducted and the performance of the ADCG network on a ship's OPT2SAR feature transfer task was evaluated using a feature visualization method.
3) In Section III-C, the image translation performance of the ADCG networks with that of several advanced and popular methods, such as U-GAT-IT [37], DualGAN [30], DiscoGAN [29], and CycleGAN [28], was compared.
4) Finally, in Section III-D, the commonly used VGG16 [7], ResNet50 [8], and MobileNet [10] were used to test the pseudo-SAR domain's ability to improve the recognition precision of popular classification networks in the SAR ATR task.

A. Experimental Data Description and Implement Details

1) Dataset: The target-domain SAR ship dataset was an open-source dataset FUSARship1.0 [38], and the optical dataset was FGSCR [39]. Fig. 6 illustrates the images from the FUSARship1.0 and FGSCR datasets.

1) *FUSARship1.0:* FUSAR-Ship is an open benchmark SAR dataset used for the development, evaluation, and benchmark testing of ship SAR ATR algorithms. It includes over 5000 high-resolution samples captured from 126 GF-3 scenes, 15 main ship categories, such as cargo ships, oil tankers, and fishing vessels, as well as 98 ship subcategories and many nonship targets. The slice size of the ship was 512 × 512 pixels.
2) *FGSCR:* FGSCR is a fine-grained ship classification optical dataset with 42 categories, consisting of 9320 optical images of the most common types of ships collected from remote sensing datasets, such as DOTA [40], HRSC2016 [41], and NWPUVHR-10 [42].

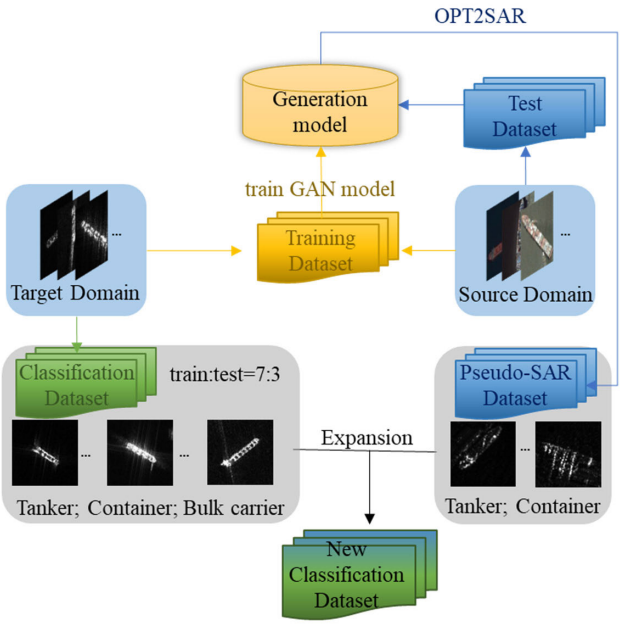


Fig. 7. Dataset partitioning diagram.

2) Dataset Partitioning: In our cross-modality domain transfer learning experiment, the dataset was divided, as shown in Fig. 7. The SAR dataset, FUSARship1.0, is the target domain training dataset, in which Tanker, ContainerShip, and Bulkcarrier are used as popular network classification datasets. The ratio of training and test data for the popular classification network was 7:3. The optical ship dataset FGSCR was used as the source domain training dataset, in which all tankers and half of the container ship were used as the generation model of pseudo-SAR test data to expand the number of Tanker and ContainerShip in the FUSARShip dataset and achieve the goal of balancing the amount of data between categories. During the training process of the ADCG network, the training and testing datasets of the optical and SAR were divided into a 9:1 ratio, and the test images were used to generate sample images to monitor whether the network experienced premature mode collapse. In the network training process, to reduce the difference in the proportion of ships in the SAR image and ships in the optical image, SAR image enhancement preprocessing was carried out, and the 512 × 512 SAR image center was cut to a size of 128 × 128 and then linearly interpolated back to 256 × 256.

3) Experimental Setup: To make sure the fairness of the ADCG network ablation experiment, all algorithm parameters were set according to the original text of CycleGAN [28]. The training epoch is set to 200, and the model is trained using adaptive motion estimation (Adam) with an initial learning rate of $lr = 0.0002$ and a decay rate of $\beta_1 = 0.5$ and $\beta_2 = 0.999$, with a momentum of 100 periods. First, the GAN network is trained and the weight model is obtained. The generation model obtained through training was then used to translate the optical images into the pseudo-SAR images.

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TABLE I
PERFORMANCE COMPARISON AND OBJECTIVE EVALUATION WITH BASELINE NETWORK

	Feature transform model		Attention module	Parameters /millions	Time /h	FID		KID	
	RBs	DBs	CBAM			$P - T_{sar}$	$T_{opt} - T_{sar}$	$P - T_{sar}$	$T_{opt} - T_{sar}$
CycleGAN	√			1.1378	10	125.95	357.04	0.0761	0.3600
DCM		√		**0.1219**	**5**	119.52		0.0733	
Att-CG	√		√	1.1444	10	118.93		0.0742	
ADCG		√	√	0.1285	5	**76.04**		**0.0403**	

The generated pseudo-SAR dataset was used to extend the dataset of the classification dataset and balance the data size difference between classes. All the experiments utilized the PyTorch framework and an NVIDIA GeForce RTX3090Ti processor.

4) Evaluation Metrics: The Fréchet inception distance (FID) [43] and Kernel inception distance (KID) [44] were used as generated image quality evaluation metrics. By calculating the multivariate normal distribution distance, the FID compares the distance of the source dataset to the target dataset at the feature level. The KID compares the feature distance between the source dataset and the target dataset by comparing the value of the square of the maximum mean difference. Different from the FID, the KID has a cubic kernel unbiased estimate that matches human perception more consistently.

a) FID: The FID extracts the 2048-dimensional vector, respectively, before the source dataset and target dataset input the fully connected layer as image features through pretrained Inception-v3 and compares the distances between real and false images at the feature level by calculating multivariate normal distribution distances. This formula is expressed as follows:

$$\mathrm{FID} = \|\mu_r - \mu_g\|^2 + T_r\left(\Sigma_r + \Sigma_g - 2\left(\Sigma_r \Sigma_g\right)^{\frac{1}{2}}\right) \tag{7}$$

where μ_r and μ_g are the mean of the features of the real image and the generated image, and Σ_r and Σ_g are the covariance matrix of the features of the real image and the generated image. When the generated image is closer to the real image features, the square of the difference between the mean and covariance is smaller, as well as the sum of 2.

b) KID: The KID is the square MMD of the polynomial kernel function in the inception feature representation space. The square MMD is expressed as follows:

$$\begin{aligned}\mathrm{MMD}^2(X, Y) = &\frac{1}{m(m-1)}\sum_{i\neq j}^{m} k\left(x_i, x_j\right) \\ &+ \frac{1}{n(n-1)}\sum_{i\neq j}^{n} k\left(y_i, y_j\right) \\ &- \frac{2}{mn}\sum_{i=1}^{m}\sum_{j=1}^{n} k\left(x_i, y_j\right)\end{aligned} \tag{8}$$

where m and n are the number of samples used to pseudo and real images, and k is the kernel. Based on (8), KID replaces x and y with a 2048-dimensional vector from the inception network, while $k(x, y) = ((1/d)x^T y + 1)^3$, where $d = 2048$ is the feature vector dimension. When the generated image features were closer to those of the real image, the KID value decreased.

B. ADCG Model Analysis

1) Ablation Experiment: Table I presents a performance comparison and objective evaluation based on the baseline network. In FID and KID, P stands for the pseudo-SAR domain image, T stands for the training dataset, and the subscripts *opt* and *sar* stand for the optical and SAR domain, respectively.

According to the objective evaluation indicators of each network in Table I, the pseudo-SAR dataset generated by each network model had significantly lower FID and KID values than the optical training dataset, indicating that each network model performed well and achieved the goal of pulling feature distances from the SAR dataset. Second, compared with the original CycleGAN network, the DBs in the DCM significantly reduced the number of network parameters, the training time halved, and exhibited excellent feature extraction performance. The improvement in the DBs solves the trouble of feature redundancy result from the stacking RBs in the original CycleGAN network with lower FID and KID values, thereby proving the effectiveness of feature extraction in DCM. Finally, the number of parameters in the CycleGAN and DCM networks with the addition of lightweight CBAM increased slightly, with a minimal increase in training time compared to those without the addition of CBAM. The decrease in the FID and KID values indicates that CBAM has a remarkable enhancement on the network model, which can suppress the negative effects of complex shore-based backgrounds and focuses on the main features of ships.

Fig. 8 shows the style translation results for the six ship images in the test dataset. First, from the diversity and clarity of the generated images, it can be seen that the generation model performs well, and the network does not experience the problem of mode collapse. Each network learns the grayscale characteristics of SAR images and the basic texture structure

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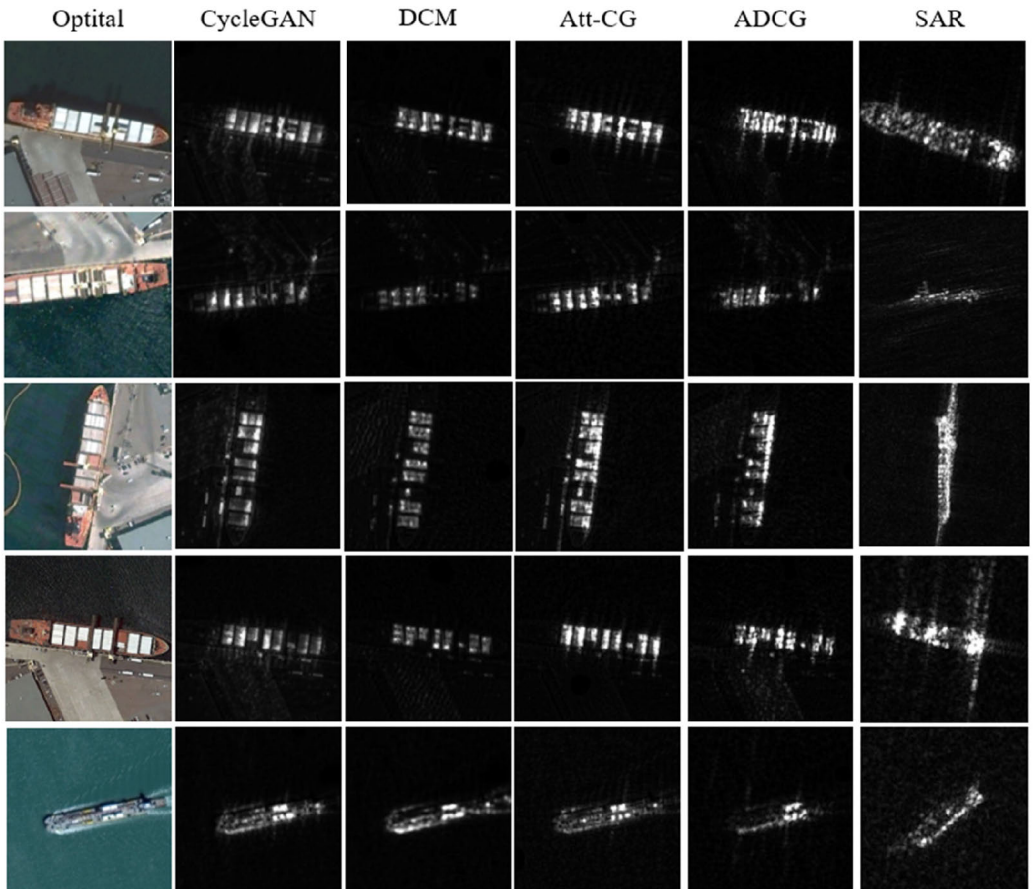


Fig. 8. Examples of pseudo-SAR images in ablation experiments.

characteristics of images and, to a certain extent, generates pseudo-SAR images with better contrast and structural details.

Second, the pseudo-SAR images generated by the DCM have higher clarity compared to the original CycleGAN model, and the main structure of the ship is more complete. Finally, the network model with added attention mechanisms has achieved the effect of suppressing complex shore-based backgrounds and focusing on the main features of the ship to a certain extent in generating images. Among them, the ADCG network model generated images that were very similar to real SAR images and even learned some of the scattering effects of SAR images.

Overall, each module proposed has a reasonable design, and the DCM has excellent feature extraction performance, solving the problem of feature redundancy caused by RBs in the baseline network and greatly reducing the number of network parameters and training time of the network model, which helps the network model locate the main features of the ship and can improve the performance of the baseline network CycleGAN in specific ship OPT2SAR image translation tasks.

2) Feature Visualization: To further validate the value of the proposed method in the cross-modality domain pixel-level feature transfer task of ship OPT2SAR, the t-distributed stochastic neighbor embedding (t-SNE) [45] method was utilized to display the feature distribution of the optical and SAR domain training sets and pseudo-SAR domain testing sets in 2-D image dimensionality reduction. The results of the feature dimensionality reduction are shown in Fig. 9. After mapping the corresponding features of each of the three domains into a 2-D space, it can be clearly seen that although there are a few purple dots representing real SAR image features, there are still optical image features intersecting these which are represented by red dots, indicating that the distribution difference between the domain of optical and the SAR images is still very significant. This indicates that cross-modality domain transfer learning in OPT2SAR is extremely difficult. However, the proposed generative adversarial method converts optical domain images into pseudo-SAR domain images across modality domains. The generated pseudo-SAR domain ship image features were highly confused with the real features, significantly reducing domain gaps and making it impossible to distinguish between the domain of the pseudo-SAR and the real SAR. This demonstrates the predominant performance of the ADCG network in OPT2SAR image translation tasks.

Owing to the high resolution of the optical images, obvious features, such as ship color and texture structure, and significant differences between different ship classes, the difficulty of differentiation was lower. Therefore, the optical images exhibit multiple feature clustering centers in the 2-D dimensionality reduction feature display results. However, the formation of the wide distribution gap in the domain of optical and SAR is not affected by the overall feature distribution in optical domain. Owing to the overall high resolution of the FUSARship dataset, there are very few high-resolution ship images in the

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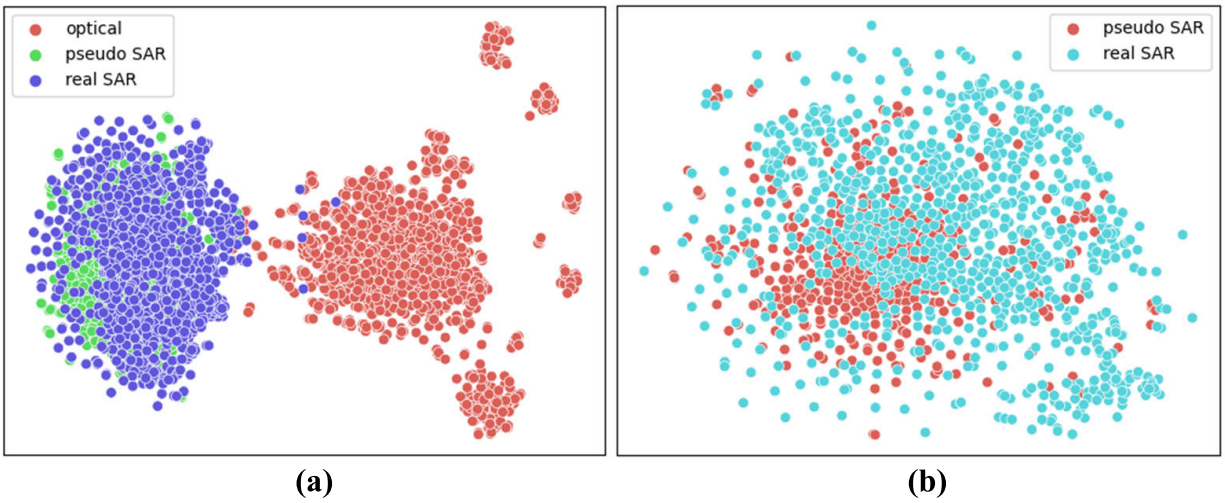


Fig. 9. 2-D reduction of t-SNE features in optical, real SAR, and pseudo-SAR datasets.

dataset with relatively complete texture structure features. This results in a slight overlap between the real SAR and the optical image features in the 2-D dimensionality reduction feature display results. However, this does not affect the differences in feature distribution between the domain of optical and SAR.

According to the feature distribution in Fig. 9(b), it can be seen that the pseudo-SAR and real SAR domains have achieved excellent results in the pixel-level feature transfer task of the OPT2SAR cross-modality domain proposed in this study. After adjusting the pixel-level image style, the pseudo-SAR domain significantly reduced the gap of feature distribution with the real SAR domain, making it almost impossible to separate it from the domain of real SAR, verifying the excellent performance of the method. Although the feature density in the pseudo-SAR domain is slightly higher than that of the real SAR domain, overall, the image features in the pseudo-SAR domain are highly confused with those in the real SAR domain, indicating that the proposed image translation model and the diversity of the pseudo-SAR domain images generated by this method perform well, and the generation model does not collapse by overfitting a single feature in the real SAR domain.

Overall, from the visualization results of the feature dimensionality reduction, the transition domain generated by the ADCG is very similar to the real SAR domain ship image features, which verifies the rationality and availability of this method in the cross-modality domain pixel-level feature transfer task of ship OPT2SAR.

C. Translation Performance Comparison

In this section, the ADCG networks were compared with CycleGAN [28], U-GAT-IT [37], DualGAN [30], and DiscoGAN [29] to verify that the proposed model achieves excellent performance in ship OPT2SAR image-style translation. CycleGAN is the baseline network for the ADCG, and the U-GAT-IT network has good transformation performance among unsupervised, unpaired methods that are currently popular and open source. DualGAN and DiscoGAN are the networks whose basic principle of image translation is similar to that of CycleGAN; however, the number of network parameters is small, and the translation speed is fast.

TABLE II
PERFORMANCE COMPARISON WITH POPULAR GENERATIVE ADVERSARIAL NETWORKS

	FID	KID	Time/h
U-GAT-IT	91.94	0.0646	18
DualGAN	128.41	0.0798	4
DiscoGAN	140.63	0.0911	**3**
CycleGAN	125.95	0.0761	10
ADCG	**76.04**	**0.0403**	5

Fig. 10 illustrates a comparison of the translation results of the six ship OPT2SAR images from each network with the real images, and Table II shows the objective evaluation indicators for each network. First, each network achieved good performance in ship OPT2SAR image translation, generating pseudo-SAR images with good clarity and diversity without any pattern collapse. Second, the U-GAT-IT network required the longest time, reaching 18 h, which is three times that of the DiscoGAN network. However, the FID and KID scores of the U-GAT-IT network reached their lowest values, except for the ADCG, which had the best effect. However, as per Fig. 10, the U-GAT-IT network did not perfectly adapt to the ship OPT2SAR task, and the complex shore-based optical background still affected the quality of the pseudo-SAR images. The DiscoGAN network required the shortest time of 3 h, and the network model was the simplest. However, from the FID and KID scores, it was observed that the quality of the DiscoGAN network image translation is not ideal, and the value is the highest; that is, the feature-pulling effect of the real SAR image is the worst. The above experimental results verify that the current popular generative adversarial networks are difficult to achieve high-quality image translation with short

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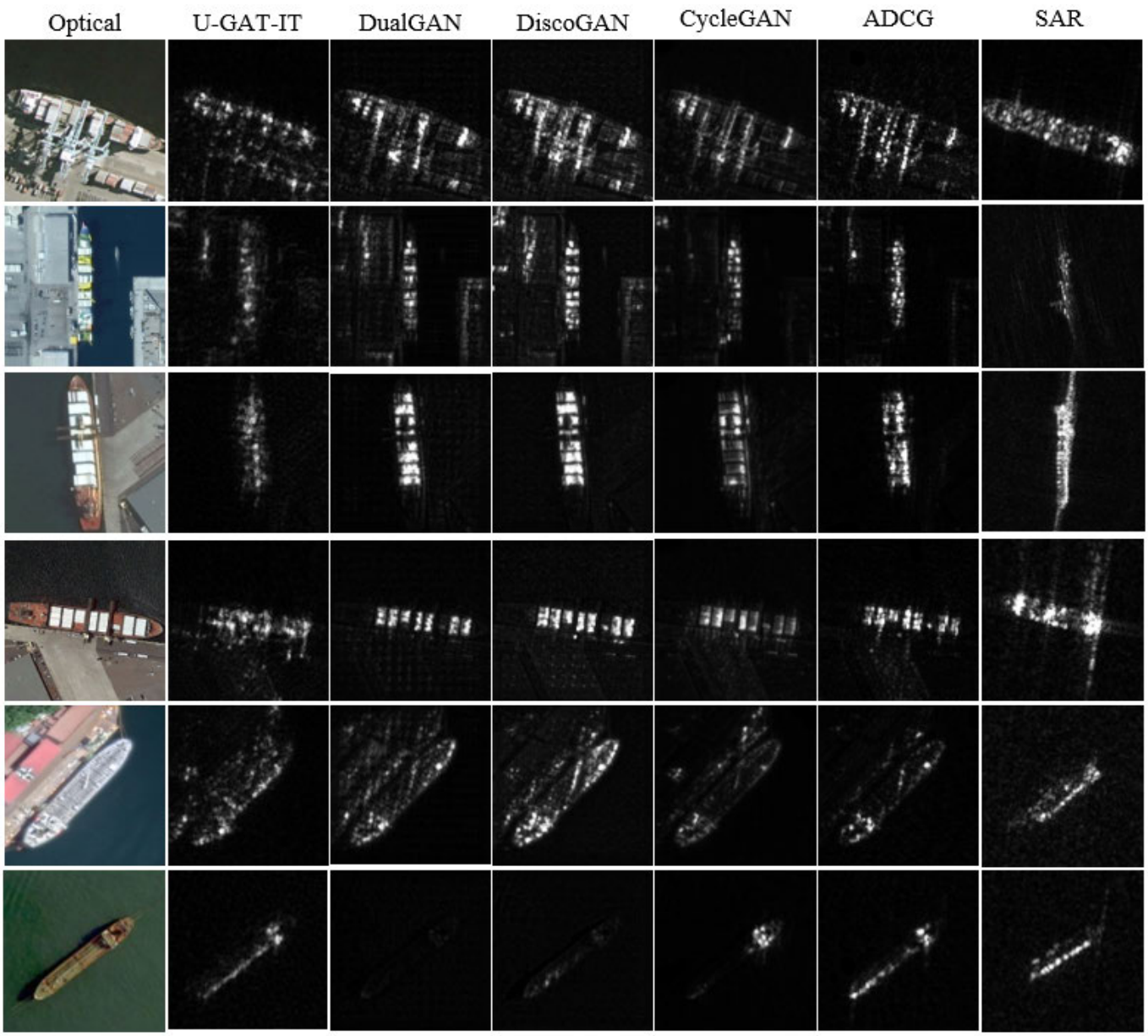


Fig. 10. Comparison of pseudo-SAR image examples in popular generative adversarial networks.

time consumption and even more difficult to eliminate the influence of complex shore-based backgrounds in ship image translation tasks.

Finally, by comparing all popular generative adversarial networks with the proposed ADCG network, the proposed network achieved the highest image translation quality with a lower model training time, and the CBAM achieved the effect of suppressed complex shore-based backgrounds, achieving a perfect fit for the ship OPT2SAR image translation task.

D. Recognition Performance Test

A pixel-level OPT2SAR feature transfer method, which aimed to solve the problem of sparsely labeled samples and imbalanced categories in the SAR dataset, was proposed to prove the rationality and availability of the pseudo-SAR domain generated by the cross-modality domain. Therefore, the ability of the pseudo-SAR to enhance the recognition precision of popular ship classification networks was tested.

A flowchart comparing the accuracy of the classification recognition model before and after expanding the dataset is shown in Fig. 11. First, the generation model training dataset is sent to the GAN for training to obtain the generation and discriminant models, and at the same time, the pictures of the popular network classification dataset are sent to the CNN for training to obtain the first prediction model. The pseudo-SAR test input data were inserted into the trained generation model, several output pseudo-SAR images were obtained, these were then merged and expanded with the training data of the real SAR network classification dataset, and the expanded dataset is input into the same CNN to obtain the second prediction model. Finally, a real SAR network classification dataset was

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TABLE III
COMPARISON OF ACCURACY FOR CLASSIFICATION NETWORKS BEFORE AND AFTER EXPANSION

		Precision			Accuracy	Mean value
		Bulkcarrier	Container	Tanker		
Before expansion	VGG16	0.87	0.50	0.88	0.74	0.79
	ResNet50	0.81	0.99	0.95	0.92	
	MobileNet	0.84	0.40	0.91	0.71	
After expansion	VGG16	0.87	0.67	0.88	0.80	**0.85**
	ResNet50	0.83	0.99	0.95	0.92	
	MobileNet	0.78	0.67	0.99	0.81	

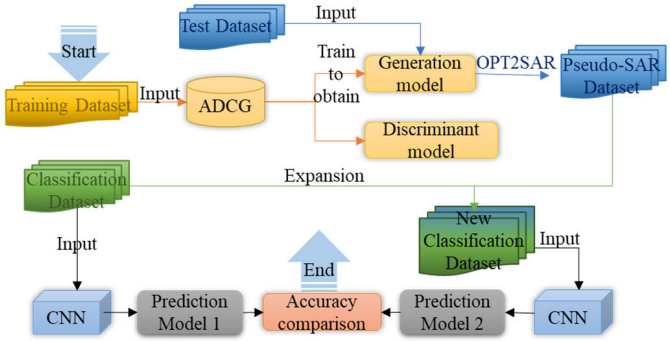


Fig. 11. Flowchart for comparing the accuracy of classification recognition models before and after expanding the dataset.

used to test the data and fed into two prediction models to compare the prediction accuracy and obtain experimental conclusions.

As shown in Table III, a pseudo-SAR domain image expansion was conducted on the container and tanker categories of the training data in the real SAR popular network classification dataset and compared the recognition accuracy of three popular classification networks, VGG16 [7], ResNet50 [8], and MobileNet [10], before and after the dataset expansion. These three networks were classified as open-source networks with good effects and high usage rates. The accuracy represents the proportion of correct predictions for all samples. The mean value is the mean of the three network accuracy rates.

It can be seen that before the expansion of the dataset, the recognition accuracy of popular classification network models was not high because of the small sample size of SAR and the extremely uneven data volume between classes. After using pseudo-SAR data samples to enlarge the dataset, the goal of improving the recognition accuracy of classification network models was achieved by balancing the differences in data volume between classes and expanding the sample data size. The three networks achieved an average improvement of 6% in the recognition accuracy. This indirectly demonstrates the rationality, effectiveness, and application value of the cross-modality-domain pixel-level feature transfer task for a ship OPT2SAR.

IV. CONCLUSION

In this study, based on the well-known unsupervised and unpaired image conversion network CycleGAN, a new network was proposed, the ADCG, that is suitable for the cross-modality domain knowledge transfer learning of ship OPT2SAR tasks. The key improvements in the ADCG are the DCM and lightweight CBAM. The DCM replaces the original RBs with DBs, which enables the ADCG to overcome the problems of generator feature redundancy and time-consuming network model training during the transfer learning task. The CBAM network can solve the problem, in which the generator cannot locate the ship's main features with a slight increase in network parameters. Moreover, the importance and effectiveness of each module through ablation experiments were demonstrated, and the ADCG demonstrated superior image translation performance with the lowest FID and KID values. The t-SNE feature visualization method also intuitively shows that the ship image features in the pseudo-SAR domain generated by the ADCG have a high degree of confusion with the ship image features in the real SAR domain, and the domain gap is significantly reduced. The performance of the ADCG network was compared with that of other popular image translation networks: U-GAT-IT [37], DualGAN [30], DiscoGAN [29], and CycleGAN [28]. The significant reduction in the training time and improvement in the image quality indicated the superior performance of the ADCG network. Finally, the commonly used VGG16 [7], ResNet50 [8], and MobileNet [10] were used to test the ability of the pseudo-SAR domain to advance the recognition precision of popular classification networks in the SAR ATR task. The average recognition accuracy of the three networks increased by 6% after the dataset was expanded. This result proves the rationality, effectiveness, and application value of the pseudo-SAR domain generated by the ADCG to overcome the trouble of scarcity and class imbalance of effective marker samples for ship SAR ATR tasks.

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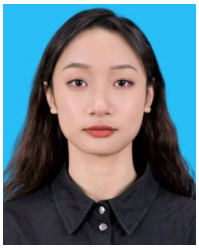
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